

Sajeer F M

AI Engineer & Software Developer

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SUMMARY

Software Developer and AI Engineer with experience building full-stack web applications and production-ready AI systems. Skilled in Python, FastAPI, and Django for backend development, with hands-on expertise in RAG pipelines, LLMs, and vector search. Passionate about solving real-world problems by combining software engineering and AI

EDUCATION

TKM College Of Engineering

Master of Computer Applications

Kollam, Kerala

Aug. 2025 – Present

University of Kerala

Bachelor of Science in Computer Science

Trivandrum, Kerala

Aug. 2021 – May 2024

PROJECTS

QueryX – AI-Powered PDF Question Answering System | [GitHub](#) | [Live Demo](#)

FastAPI, HTML, CSS, JS, Qdrant, Gemini API, Docker, Google Cloud

Jan 2026 – Mar 2026

- Built a **RAG-based PDF Q&A system** where users upload documents and ask questions in natural language, receiving accurate answers strictly from the document content
- Implemented **Hybrid Search** combining **BM25** and **Vector Search with Qdrant** to retrieve the most relevant document chunks before passing context to the LLM
- Integrated **Gemini API** to generate document-grounded responses, minimizing hallucinations by conditioning answers strictly on retrieved content
- Deployed on **Google Cloud** using **Docker** and **Nginx** with **HTTPS** support via reverse proxy configuration

Exam Seat Allocation System | [GitHub](#)

Django, Python, PostgreSQL, Hyper-Heuristic Optimization, HTML, CSS, JS

Jul 2026 – Present

- Built a **web-based exam seat allocation system** using Django to automate seating arrangements for large-scale examinations, eliminating manual scheduling effort
- Implemented a **hyper-heuristic optimization algorithm** that dynamically selects and applies heuristic strategies to generate **conflict-free seating plans** across multiple halls
- Enforced institutional constraints including **hall capacity limits, department separation, subject overlap, and consecutive seat restrictions** during allocation
- Built modules for **student registration, hall management, course scheduling, and admin dashboard** with secure role-based access control
- Integrated **PostgreSQL** for data management and generated **downloadable seating charts and allocation reports** for examination administration

TECHNICAL SKILLS

Programming Languages: Python, C, SQL, JavaScript, HTML/CSS

AI/ML: LangChain, LangGraph, RAG, LLMs, Vector Embedding, ChromaDB, Qdrant

Frameworks: FastAPI, Django, SQLAlchemy

Databases: MySQL, PostgreSQL, SQLite

Developer Tools: Git, GitHub, Docker, AWS, Postman, N8N

CERTIFICATIONS

Python Developer (Google) | Data Science (Cisco) | Web Development (freeCodeCamp) | Generative AI (IBM)